Case study

1. Setup

Every controller runs in every situation 100 times, with the seeds 1 to 100. All controllers use the same seeds, so they have the same random numbers. A run has 400 steps. The six situations are: S1 is the start of the program, S2 is an unknown inflow change (u_0 goes from 50 to 62), S3 is a set point change (from 74 to 20), S4 is heavy rain and then strong evaporation, S5 is S2 with noise on the level reading, and S6 is an extreme change of the flows (u_0 goes from 50 to 75).

The weights of the controllers come from the tuning done in modeling of the experimental part:

Controller	Weights
P	K_P= 6
PI	K_P= 3, K_i= 0.2
PID	K_P= 14, K_i= 0.02, K_d= 100
PID+	K_P= 4, K_i= 0.2, K_d= 10
Ours	K_P= 10, K_i= 0.2, K_d= 25, k = 0.1 for f_p, 0.01 for f_i, 0 for f_d, dead-band 0.2

We use the metrics from chapter 6 of the modeling of the experimental part. A smaller IAE means a smaller total error.

2. E1: comparison with P, PI, PID and PID+

Table 4.1 shows the mean IAE of 100 runs, with the standard deviation in brackets.

IAE mean (std)	S1	S2	S3	S4	S5	S6
P	1444 (124)	2450 (125)	2244 (142)	2014 (281)	2474 (118)	3665 (144)
PI	1843 (76)	1963 (76)	2768 (102)	2476 (230)	2015 (76)	2272 (167)
PID	1441 (76)	1394 (90)	1897 (90)	1942 (234)	1633 (107)	1919 (110)
PID+	1142 (147)	1262 (147)	1776 (180)	1733 (193)	1379 (139)	1588 (165)
Ours	1083 (104)	1270 (102)	1631 (130)	1560 (206)	1606 (98)	1757 (118)

Table 4.1. The IAE of the five controllers in the six situations (E1)

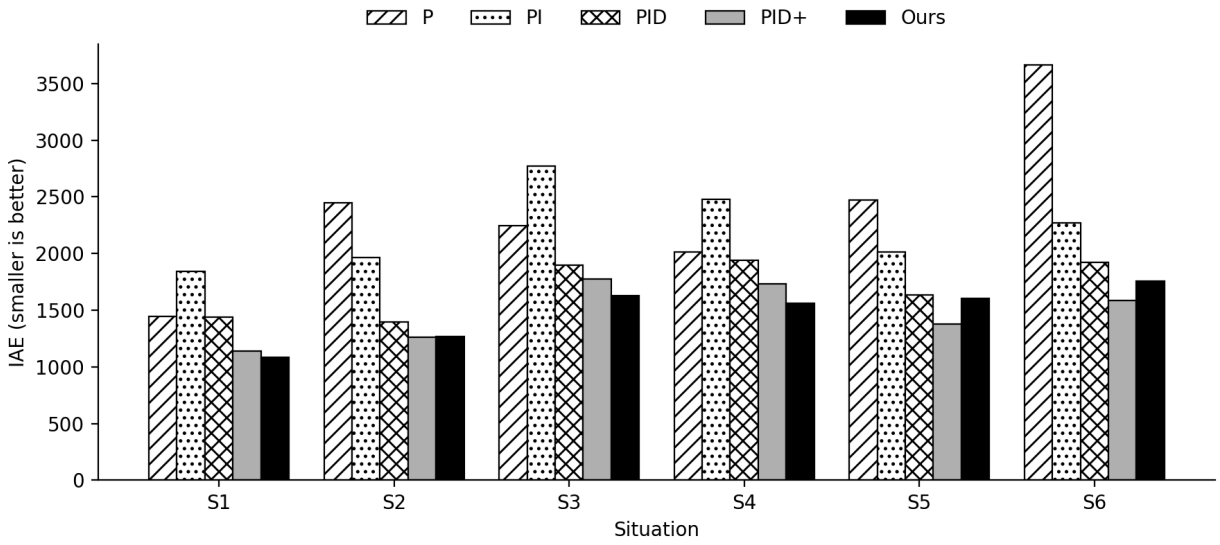


Figure 4.1. The IAE of the five controllers in the six situations (E1)

Table 4.2 shows in how many runs (percent) the level stays under 1 percent from the set point until the end, and Table 4.3 shows the valve movement.

Settled runs (%)	S1	S2	S3	S4	S5	S6
P	100	0	100	100	0	0
PI	100	100	100	100	100	100
PID	63	73	82	56	41	17
PID+	100	100	100	100	96	100
Ours	100	81	100	100	22	75

Table 4.2. The percent of settled runs (E1)

Valve movement	S1	S2	S3	S4	S5	S6
P	164	188	303	439	772	254
PI	119	139	221	328	479	203
PID	1130	1127	1131	1138	1158	1131
PID+	100	136	191	390	900	180
Ours	154	258	278	533	1047	275

Table 4.3. The valve movement (the sum of the changes of the valve position) (E1)